

Active Inference ~ Parr, Pezzulo, Friston ~ Chapter 4

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chapter 4 is called the generative models of active inference and it begins with a quotation everything should be made as simple as possible but not simpler by Albert Einstein Ali what is your overview thought or warning for chapter four okay so uh after the preliminary materials um in chapters two and three uh which was basically largely based on um concept providing some conceptual framework for developing uh the further Theory chapter four delves into much more detail in terms of mathematical formulation and it unpacks a lot more uh the way that the central equations of active inference is derived and how to construct the important elements of active inference models so say matrices a b c and d and also how to put together a generative models in different situations so it basically lays out the foundation or um for I mean constructing an active inference models uh both for discrete time situations and continuous time at once uh which will be used later in chapters seven and eight but this is uh probably one of the most uh challenging and uh at least mathematically dense chapters in the book so I would personally suggest reading through this chapter really slowly and even if we don't get to understand every single detail of the chapter uh obviously we can return required as we go through uh the textbook thank you Ali yes so let's look through the sections just to add on though chapter four is one of the larger and more equation dense chapters because it is the common kernel or basis that's then going to get applied in chapter five in the neurobiological case there's a recipe for making chapter 4 in chapter six that's the recipe for active inference modeling chapter 7 and 8 are about the discrete and the continuous time variant or or subtype or motif of these kinds of things called generative models so this is the real common root and we'll just look at what the sections are this chapter complements the preceding chapters conceptual treatment of active inference with a more formal treatment section 4.2 from Bayesian inference to free energy what would you say about this sectionally okay so as we know the free energy principle is inspired by previous work on Bayesian

inference I mean all the way way back to helmholtz theory about um unconscious inference or something to that effect I can remember the exact term but here um I mean it provides in a more in a bit more detail how we can derive pre-energy principle uh formalism using the established Bayesian inference formulation and particularly one of the key uh movements or at least one of the key decisions uh in uh through the derivation of free energy principle formulation is using uh Jane's inequality principle to derive uh an upper upper bound instead of just using uh the exact values to compute um or or to achieve required parameters so uh that's basically uh in my opinion the key premise of uh section 4.2 and to see uh in a bit more detail how we can achieve those uh upper bounds using uh Jane's inequality uh directly from by using I mean manipulations of Bayesian inference based in statistical formalism thanks I'll just add one point from this section broadly these are the problems of inferring states of the world perception and inferring a course of action planning so this is again referring to the perception and action and everything that happens in between is the internal or the cognitive part of the inference but this is like the blanket State cybernetic input output and then let's look at the first equation or how much equations overall or what equations do you think we should highlight uh okay so um maybe we can uh I mean uh just uh as a general comment about these um different equations uh well each of these equations provide a distinct step toward deriving uh the ultimate whole picture so uh even if we don't quite understand how we can derive from uh each step to the other one it's good to know that it's only required to understand how we get to that ultimate whole picture but ultimately what we would need in order to develop um for a develop active inference models is the ultimate equation or ultimate whole picture so this is just a way to elucidate uh the steps toward developing uh that whole picture but um again it's not an essential requirement to understand um the materials of the rest of the book uh but if we go from uh I mean equations 4.1 toward the 4.4 or other words the free A variation of free energy well equation 4.1 is just a basic definition of um probably some properties of probabilities in terms of conditional probability and so on so equation 4.2 provides the central chains inequality principle and how it relates to um I mean conditional probabilities and uh of course joint probabilities and then by using uh those uh two properties or those two equations we ultimately get to 4.4 which is the definition of variation free energy parameter uh which is the parameter of interest that needs to be optimized in order to inference to happen or at least perceptual inference to happen in active inference models thanks only thing all add is f is the letter used for variational free energy you can think of it like a computer program and the arguments that it takes in or the variables that it takes in are Q which is the distribution that's

under the statisticians control and Y which are the data which are outside of the statisticians control do you want to describe more about anything in this equation or carry on uh just one thing to uh that can probably be helpful is to somehow compare these steps with um the initial picture we had from uh chapter two uh because uh variation of free energy was first introduced in chapter two so uh it can be helpful to go back and forth between chapters two and four and try to connect the dots between um the related points there section 4.3 generative models all right I'll read the first sentence then you can give some thoughts to calculate the for the free energy we need three things data a family a variational distributions and a generative Model comprising A prior and a likelihood in this section we outline two very general sorts of generative model used for active inference and the form the free energy takes in relation to each but okay so uh as mentioned earlier this chapter deals both with discrete time and continuous time situations uh so clearly we would need two different types of generative models for each situation obviously the the generative models or the way to construct generative models for discrete time situations uh would vary quite a bit from uh the one for continuous time situations but the general principle underlying uh those generative models are basically the same which is uh I mean to somehow construct a model of the environment and I mean either be it for the situation that that is sequential in time or for the situations that need to be somehow each moment of the situation needs to be accommodated in terms of a continuous time situation so uh figure 4.2 provides some examples of both so for uh yes let me see yes so we have some examples of different kinds of generative models some uh case case studies if you like and it provides various ways to show how the the dependencies between between variables can be modeled using these kinds of graphical probabilistic models so one common way to represent General to models is to use these kinds of graphical probabilistic models in active imprints literature which is at least in this case the circles would represent the random variables and the squares that would represent the distributions which would describe the dependencies between those random variables so we can see uh the clear relationships between those parameters here which is basically what this whole graph what constitutes the generative model that needs to be used for different situations and then in figure 4.3 we can compare the two different types of generative models based on uh whether it's discrete time or continuous time situations so the upper picture is a generative model for the discrete time situation and the lower picture is the parallel continuous time version of it and as we can see the general topology of these models are the same the only things that differ is the use of parameters for policies or I mean discrete time policies or the continuous templates and we

can obviously compare the different elements for for both uh priors States external States internal States and so on by comparing these two models here yeah we often return to figure 4.3 it's kind of the Rosetta Stone of generative modeling for the context of this book because it's then going to develop out into chapter seven and eight and it represents a really fundamental decision made in modeling in the later chapters it's also shown how it can be made into a hierarchical model that combines aspects of both but within each level of modeling still these are the kinds of decisions that modelers are presented with when it comes to statistical modeling overall so section 4.4 goes into essentially the top half of figure 4.3 discrete time what would you say about discrete time okay so uh the discrete time situation uh is uh obviously the archetype discrete time situation which is upon the Palm DP models so at this point I would very much like to recommend uh following uh the steps following the material from step by step paper because uh that's uh in that paper uh the way to construct pump DP models is described in a bit more detail so if anyone feels like they should learn a bit more about the the gaps in the details uh I I would very much like to recommend that particular paper so uh yeah uh I don't know how much detail uh we should go into because uh it's I mean uh although uh it's not maybe a detailed enough for some tastes but it goes in a quite uh extensive detail uh about how we can construct these models using the the concepts we've learned uh in previous chapters uh so ultimately we reach equations 4.13 and 4.14 which are basically the culmination of palm DP formulation using the vector notations and gradients and so on so uh that's great and we go to continuous time situation great few things intervene in the continuous time chapter ones that we'll just mention here because they're kind of boxed or partitioned from The Continuous time part but they're following Pages first is Markov blankets we won't go into it here but kind of footnote that or look at some other places where we talk about it outside of this chapter overview figure 4.4 AZ message passing again a big topic let's kind of just go past it now back to the regularly scheduled continuous time generative model discussion and then another box to the generalized coordinates of motion so taking position plus derivatives of position and that has some beneficial properties that are described and unpacked also elsewhere do you want to say anything about 4.5.2 uh well the only thing that comes to mind is although as I said before all the formulations here may look uh more I mean a bit too dense to understand at the first pass but uh some of the key maybe components here could be uh obviously the the material from box 4.2 and 4.3 uh is uh I think are quite essential to understand uh the uh the underlying principle behind uh deriving uh The Continuous time situation because without LaPlace approximation what we would have uh in

terms of free energy minimization is uh it would look very much like the Gibbs it gives free energy so uh I mean the key distinction between uh the free energy Principle as described in active inference literature as opposed to Gibbs free energy is this distinction uh is the LaPlace approximation so this is what enables us to go from Gibbs free energy to um I mean the variation of free energy so uh yeah that's uh I mean quite essential to uh make this to be familiar with this essential approximation and obviously the concept of generalized coordinates of motion will come time and time again throughout the whole book particularly in chapters uh eight and nine uh so yeah those two concepts I believe needs a bit more attention so yeah sounds good box 4.3 LaPlace approximation equations another message passing representation and a summary the key message to take away is that approximate Bayesian inference may be framed as minimizing a quantity known as variational free energy this depends on the generative model that expresses our belief about how data are generated anything else you want to add ah nothing comes to mind at the moment because as I said this is I mean we're still in the stage that we want to develop are essential tools to be used in the rest of the book so uh here up until now I Believe by the end of chapter four we have acquired all the essential necessary mathematical tools and the next chapter chapter 5 is kind of acts like an interlude and I don't think it's the direct um I mean continuation of chapters uh one through four so I believe the the first um section or the first part of the book uh conceptually and mathematically ends here so yeah that's it yes it's a little bit like the pragmatic modeling part gets foreshadowed or explored in five now that we're all built up with four all right that's the end of the overview for four